American Options and Optimal Stopping

Lecture 10 Notes — Exercise Rights, Dynamic Programming, and Free Boundaries

Goals and logical structure

European claims have one prescribed cash-flow date. An American claim lets its holder choose a stopping time, so its price combines no-arbitrage valuation with a timing decision:

$$\begin{aligned}\text{exercise right} &\longrightarrow \text{optimal stopping under } \mathbb{Q}\\ &\longrightarrow \text{backward induction / Snell envelope}\\ &\longrightarrow \text{obstacle PDE and numerical methods.}\end{aligned}$$

At every date ask one question: is immediate exercise worth more than continuation?

1 The exercise right changes the pricing problem

1.1 European and American exercise

Definition 1.1 (European and American claims)**.** Let g_t be the cash payoff received if a claim is exercised at date t. A *European* claim pays g_T at one fixed maturity T. An *American* claim may be exercised at any stopping time

$$\tau \in \mathcal{T}_{t,T} = \{\tau : \tau \text{ is a stopping time and } t \le \tau \le T\}.$$

Its cash flow is g_τ.

For a stock S and strike K, the exercise payoffs are

$$g^{\text{call}}(S_t) = (S_t - K)^+, \qquad g^{\text{put}}(S_t) = (K - S_t)^+.$$

The word American describes the exercise convention. It does not describe the stock or the exchange.

Example — more rights cannot reduce value

The holder of an American put can choose $\tau = T$, reproducing the European exercise policy. Hence

$$V_t^{\text{Am}} \ge V_t^{\text{Eur}}.$$

The difference is the *early-exercise premium*; it can be zero even though the American holder owns an additional right.

1.2 The seller's super-hedging viewpoint

Let $B_t = e^{rt}$ be the bank account and X^π the wealth of a self-financing strategy. The writer must be ready whenever the holder exercises. Thus π is a super-hedge when

$$X_u^\pi \ge g_u \quad \text{for all } u \in [t, T],$$

equivalently $X_\tau^\pi \ge g_\tau$ for every admissible stopping time.

Two equivalent views in a complete market

$$V_t^{\mathrm{Am}} = \inf_{\pi}\{X_t^{\pi} : X_u^{\pi} \geq g_u \ \forall u \in [t,T]\} = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}[e^{-r(\tau-t)} g_\tau \mid \mathcal{F}_t].$$

The first formula protects the writer; the second describes the holder's optimal decision. Completeness is the bridge between them.

A stopping time is essential

The holder cannot select an exercise date after seeing future prices. The exercise decision at date t may use only $\mathcal{F}_t$. A future-dependent choice would overstate the contract value.

Exercise — first comparison

Which inequality always holds between American and European options with the same payoff and maturity? Does it mean that exercise is optimal at time zero?

Short solution

$V_0^{\mathrm{Am}} \geq V_0^{\mathrm{Eur}}$, because waiting to maturity is available. It does not identify the optimal policy: the holder may continue at every date before maturity, in which case the two values coincide.

2 Discrete time: backward induction

2.1 Risk-neutral recursion

Consider dates $0, 1, \ldots, N$, a bank account growing by $R = 1 + r > 0$, and an equivalent martingale measure $\mathbb{Q}$. Let $Y_n = g_n(S_n)$. The American value is

$$V_n = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{n,N}} \mathbb{E}^{\mathbb{Q}}[R^{-(\tau-n)} Y_\tau \mid \mathcal{F}_n].$$

At maturity there is no remaining choice: $V_N = Y_N$.

Theorem 2.1 (Bellman recursion)**.**

$$V_n = \max\left\{Y_n, \frac{1}{R}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]\right\}, \qquad V_N = Y_N.$$

Proof. Any exercise policy either stops now, giving Y_n, or continues for at least one period. In the latter case the best remaining value is $R^{-1}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]$. Take the better alternative. □

Exercise versus continuation

Define $E_n = Y_n$ and

$$C_n = \frac{1}{R}\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n].$$

Then $E_n > C_n$ means exercise, $E_n < C_n$ means continue, and equality means that either action is optimal.

2.2 A binomial implementation

In a binomial model

$$S_{n+1} = \begin{cases} uS_n, & \text{up}, \\ dS_n, & \text{down}, \end{cases} \qquad d < R < u,$$

the risk-neutral up probability is $p = (R-d)/(u-d)$. Thus

$$C_n = \frac{pV_{n+1}^u + (1-p)V_{n+1}^d}{R}, \qquad V_n = \max\{g(S_n), C_n\}.$$

Work from terminal payoffs backward through the tree, recording nodes where the intrinsic value wins.

Example — a strict early-exercise node

Take $S_0 = 100$, $u = 1.2$, $d = 0.8$, $R = 1.05$, and a put with $K = 105$. Then $p = 0.625$. At a date-1 down node $S_1 = 80$, exercise pays 25. The two terminal put values are 9 and 41, so

$$C_1(80) = \frac{0.625(9) + 0.375(41)}{1.05} = 20.$$

Since $25 > 20$, early exercise is strictly optimal at this node.

The up-node value is $V_1(120) = 3.214$, hence

$$V_0^{\mathrm{Am}} = \frac{0.625(3.214) + 0.375(25)}{1.05} \approx 10.84.$$

The European value is approximately 9.06, leaving an early-exercise premium of about 1.78.

Exercise — one backward step

Let $R = 1.02$, $p = 0.6$, and $K = 100$. At a node with $S = 85$, an American put has terminal continuation values 4 and 22. Exercise, continue, or indifferent?

Short solution

Exercise pays 15, while continuation is

$$\frac{0.6(4) + 0.4(22)}{1.02} \approx 10.98.$$

The holder exercises. Use terminal option values, not terminal stock prices, in the continuation calculation.

2.3 Snell envelope and the hedge

Definition 2.2 (Discounted Snell envelope)**.** With $Z_n = R^{-n} Y_n$, define

$$U_n = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{n,N}} \mathbb{E}^{\mathbb{Q}}[Z_\tau \mid \mathcal{F}_n].$$

U is the smallest $\mathbb{Q}$-supermartingale dominating Z, and $V_n = R^n U_n$.

Optimal policy

One optimal stopping rule is the first entry into the exercise region:

$$\tau_n^\star = \inf\{k \in \{n, \dots, N\} : V_k = Y_k\}.$$

In a complete market, the martingale part of the Snell envelope yields the dynamic hedge for the seller.

3 Continuous time: stopping regions and boundaries

3.1 Black–Scholes formulation

Under $\mathbb{Q}$, suppose

$$\frac{\mathrm{d}S_t}{S_t} = (r - q)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad \sigma > 0.$$

For exercise payoff $g(S_t)$, the Markov value is

$$\boxed{v(t,s) = \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}_{t,s}^{\mathbb{Q}}[e^{-r(\tau - t)} g(S_\tau)].}$$

The continuation and exercise regions are respectively where $v(t,s) > g(s)$ and $v(t,s) = g(s)$.

3.2 A non-dividend-paying call

Proposition 3.1 (No early exercise for a standard call)**.** If $q = 0$ and $r \geq 0$, then

$$C^{\mathrm{Am}}(t,s) = C^{\mathrm{Eur}}(t,s)$$

for a call payoff $(s-K)^+$. Early exercise is never strictly optimal before maturity.

Proof. For $\theta = T - t > 0$,

$$C^{\mathrm{Eur}}(t,s) \geq e^{-r\theta}\mathbb{E}^{\mathbb{Q}}_{t,s}[S_T - K] = s - Ke^{-r\theta} \geq s - K.$$

Together with nonnegativity, this gives $C^{\mathrm{Eur}}(t,s) \geq (s-K)^+$. At any stopping time, retaining the European call is at least as valuable as exercising it, and its discounted price is a martingale. □

The dividend qualification matters

When $q > 0$, holding the stock before an ex-dividend date can be valuable. An American call may then be exercised early. Never quote the preceding result without the no-dividend assumption.

3.3 American puts and boundaries

For a put, receiving $K - S_t$ now can be attractive at low stock values. In the standard model with $r > 0$, the exercise region typically has the form

$$\mathcal{E} = \{(t,s) : s \leq b(t)\}, \qquad \mathcal{C} = \{(t,s) : s > b(t)\},$$

where $b(t) < K$ before maturity. Deep below K, waiting delays cash while adding little useful optionality; near the strike, continuation is valuable.

4 The obstacle PDE

4.1 Variational inequality and free boundary

Let

$$\mathcal{L}f = (r-q)sf_s + \frac{1}{2}\sigma^2 s^2 f_{ss}.$$

In the continuation region, hedging gives

$$v_t + \mathcal{L}v - rv = 0.$$

Together with the requirement $v \geq g$, this becomes

$$\boxed{\min\{v - g, -(v_t + \mathcal{L}v - rv)\} = 0, \qquad v(T,s) = g(s).}$$

For a regular American put boundary $s = b(t)$, value matching and smooth fit read

$$v(t,b(t)) = K - b(t), \qquad v_s(t,b(t)) = -1.$$

The boundary is free: it is a result of the pricing problem, not a contract input.

4.2 A complete PDE problem for the American put

For the numerical example below, specialize to no dividends $q = 0$, $r \geq 0$, and $g(s) = (K-s)^+$. Define the pricing operator, including discounting, by

$$\mathcal{A}f = rsf_s + \tfrac{1}{2}\sigma^2 s^2 f_{ss} - rf.$$

The variational inequality is equivalent, wherever the derivatives exist, to

$$\boxed{v - g \geq 0, \qquad -(v_t + \mathcal{A}v) \geq 0, \qquad (v-g)(v_t + \mathcal{A}v) = 0.}$$

In the continuation region $v > g$, the last condition forces the PDE equality. In the exercise region $v = g$, the PDE residual need not vanish. The terminal and spatial conditions are

$$v(T, s) = (K - s)^+, \qquad v(t, 0) = K, \qquad \lim_{s\to\infty} v(t, s) = 0.$$

At zero stock price, immediate exercise pays K; postponing that payment cannot improve its discounted value when $r \geq 0$. On a numerical domain $[0, S_{\max}]$, with $S_{\max} > K$, replace the condition at infinity by $v(t, S_{\max}) \approx 0$, and check the truncation error by increasing $S_{\max}$.

An exercise grid is an approximation to continuous exercise

For exercise restricted to $0, h, 2h, \ldots, T$, the Bermudan value satisfies

$$v_h(t, s) = \max\{g(s), \mathbb{E}^{\mathbb{Q}}_{t,s}[e^{-rh} v_h(t + h, S_{t+h})]\}.$$

For a finite h, this forbids exercise between consecutive grid dates. The continuous-time American value is recovered as the mesh tends to zero; the same finite-step identity is not exact for unrestricted exercise.

Exercise — identify the region

For an American put, a numerical scheme reports $g(s) = 18$ and a continuation estimate $C = 18.7$. Which region contains the node and what is the option value?

Short solution

It is in the continuation region and $v = 18.7$. Intrinsic value is a lower obstacle, not necessarily the price.

5 Computing American prices

5.1 Finite differences: grid and explicit update

Keep the no-dividend American put assumptions above. Choose

$$s_i = i\Delta s, \quad \Delta s = S_{\max}/M, \qquad t_n = n\Delta t, \quad \Delta t = T/N,$$

and write $V_i^n \approx v(t_n, s_i)$, $g_i = (K - s_i)^+$. Initialize $V_i^N = g_i$, fix $V_0^n = K$, $V_M^n = 0$, and march backward from $n = N - 1$ to 0. At an interior stock node, use

$$v_s(t_n, s_i) \approx \frac{V_{i+1}^n - V_{i-1}^n}{2\Delta s}, \qquad v_{ss}(t_n, s_i) \approx \frac{V_{i+1}^n - 2V_i^n + V_{i-1}^n}{(\Delta s)^2}.$$

Let $\mathcal{A}_h$ be the resulting discrete pricing operator. An explicit continuation step evaluates it at the already known level:

$$\frac{V_i^{n+1} - C_i^n}{\Delta t} + (\mathcal{A}_h V^{n+1})_i = 0.$$

Since $s_i = i\Delta s$, this gives the implementable update

$$\boxed{C_i^n = a_i V_{i-1}^{n+1} + b_i V_i^{n+1} + c_i V_{i+1}^{n+1}, \qquad V_i^n = \max\{g_i, C_i^n\},}$$

where

$$a_i = \frac{\Delta t}{2}(\sigma^2 i^2 - ri), \qquad b_i = 1 - \Delta t(\sigma^2 i^2 + r), \qquad c_i = \frac{\Delta t}{2}(\sigma^2 i^2 + ri).$$

Thus each new value uses three neighbouring values at the next time level, followed by the exercise test. Compute all interior nodes from the old array before replacing it. The weights sum to $1 - r\Delta t$, the first-order approximation to the one-step discount factor. At the end, interpolate V^0 to S_0 if necessary. Nodes with $V_i^n = g_i > 0$ identify the put exercise region; its largest stock node approximates the free boundary.

Positivity and the explicit time-step restriction

Require $a_i, b_i, c_i \geq 0$ for all $1 \leq i \leq M-1$. With $r \geq 0$, sufficient conditions for the centered stencil are

$$\sigma^2 i^2 \geq ri \quad \text{at every interior node,} \qquad \Delta t \leq \frac{1}{\sigma^2(M-1)^2 + r}.$$

Then the continuation update is monotone and its nonnegative weights sum to at most one; taking the maximum with the payoff preserves monotonicity. Reducing Δt fixes a negative b_i, but cannot fix a negative a_i. If the drift dominates diffusion near zero, replace the centered drift there by a monotone one-sided approximation and recompute the weights. For $r \geq 0$, the forward difference $rs_i(V_{i+1} - V_i)/\Delta s$ gives nonnegative off-diagonal weights.

Example — one step before maturity

Take $K = 100$, $r = 5\%$, $\sigma = 30\%$, $\Delta s = 5$, $\Delta t = 1/252$, and $S_{\max} = 200$, so $M = 40$. These parameters satisfy the positivity and time-step conditions. Starting from the terminal payoff, two nodes give

s_i	$(V_{i-1}^N, V_i^N, V_{i+1}^N)$	C_i^{N-1}	g_i	Action
80	$(25, 20, 15)$	19.9802	20	Exercise
100	$(5, 0, 0)$	0.3472	0	Continue

At $s = 80$, the payoff is locally linear and $C = 20 - rK\Delta t = 19.9801587$. At $s = 100$, $a_{20} = \frac{1}{2 \cdot 252}(36-1) = 0.0694444$, giving $C = 5a_{20} = 0.3472222$. These are numerical grid values, not exact continuous-time prices.

5.2 Implicit Euler and discrete complementarity

To avoid the explicit time-step restriction, evaluate spatial terms at the unknown level n. With the same a_i, c_i, set

$$d_i = 1 + \Delta t(\sigma^2 i^2 + r), \qquad R_i^n = d_i V_i^n - a_i V_{i-1}^n - c_i V_{i+1}^n - V_i^{n+1}.$$

The fully implicit obstacle problem is

$$\boxed{V_i^n - g_i \geq 0, \qquad R_i^n \geq 0, \qquad (V_i^n - g_i)R_i^n = 0, \quad i = 1, \ldots, M-1.}$$

The endpoint values are fixed at K and 0. The residual is $-\Delta t$ times the discrete PDE residual, explaining its sign. Provided $a_i, c_i \geq 0$, the tridiagonal system has positive diagonal, nonpositive off-diagonals and strict diagonal dominance: $d_i - a_i - c_i = 1 + r\Delta t > 0$. The explicit restriction on Δt is no longer needed for monotonicity, although accuracy still requires time and space refinement.

An implementable projected Gauss–Seidel solve

At each time level, initialize $U_i^{(0)} = V_i^{n+1}$, keep $U_0^{(k)} = K$, $U_M^{(k)} = 0$, and sweep $i = 1, \ldots, M-1$:

$$U_i^{(k+1)} = \max\left\{ g_i, \frac{V_i^{n+1} + a_i U_{i-1}^{(k+1)} + c_i U_{i+1}^{(k)}}{d_i} \right\}.$$

Repeat the sweeps until

$$\max_{1 \leq i \leq M-1} \left| \min\{U_i^{(k)} - g_i,\ R_i(U^{(k)})\} \right| \leq \varepsilon,$$

then accept $V^n = U^{(k)}$ and move to the preceding time level. Here $R_i(U) = d_i U_i - a_i U_{i-1} - c_i U_{i+1} - V_i^{n+1}$.

One unconstrained implicit linear solve followed by a single pointwise maximum does not generally solve this coupled complementarity problem: changing one node changes its neighbours' PDE residuals. That procedure is a splitting approximation; projected iteration enforces both conditions together. Policy iteration and penalty methods provide other solvers.

Further reading: P. Forsyth, *Numerical PDE Option Pricing*, CS 774, Assignment 2 (2009), for the complementarity formulation and implicit treatment of the obstacle.

5.3 Trees and least-squares Monte Carlo

A recombining tree applies the Bellman maximum at each node and makes the exercise region visible. Finite-difference schemes solve a discrete obstacle problem: each grid value must satisfy both a PDE relation and $v \geq g$. Projection, penalty, and policy-iteration methods enforce this constraint.

For many risk factors, Longstaff–Schwartz approximates continuation values from simulated risk-neutral paths:

1. simulate paths at a set of exercise dates;
2. set terminal cash flows to $g(S_T)$;
3. work backward and regress discounted future cash flows on basis functions of the current state, using in-the-money paths;
4. exercise when $g(S_{t_n})$ exceeds the fitted continuation value;
5. average the discounted cash flows induced by the estimated policy.

What LSM estimates

Regression estimates the conditional continuation value

$$C_n(s) \approx \mathbb{E}^{\mathbb{Q}}[e^{-r(t_{n+1}-t_n)}V_{n+1} \mid S_{t_n} = s],$$

not the price directly. Poor basis functions, sparse exercise dates, or too few paths can produce a poor stopping policy.

Mandatory numerical checks

Refine the time grid and verify

$$V^{\mathrm{Am}} \geq V^{\mathrm{Eur}} \geq 0, \qquad V^{\mathrm{Am,put}} \geq (K - S_0)^+.$$

For a non-dividend-paying call, the American method should converge to the European benchmark. A material gap is a diagnostic warning.

Exercise — diagnose two numerical errors

You price an up-and-in American put by Monte Carlo with daily monitoring. Name one issue caused by the barrier and one caused by the American feature.

Short solution

A path may cross the barrier between monitoring dates, creating monitoring bias; refine the grid or use a bridge correction. The American feature requires a stopping policy, so test exercise dates, regression bases, and out-of-sample policy performance.

6 Synthesis and guided practice

A practical workflow

1. Write the payoff and permitted exercise dates.
2. Compare exercise and risk-neutral continuation values.
3. In a tree, apply the Bellman maximum from maturity backward.
4. In continuous time, locate stopping and continuation regions.
5. Check numerical output against bounds and refinements.

Exercise — final stopping-policy check

An American put has immediate exercise value 12 and continuation value 11.5. A colleague adds them and reports an option value of 23.5. Explain the error, state the correct value, and give the action.

Short solution

Exercise and continuation are mutually exclusive policies, so the recursion takes their maximum:

$$V = \max\{12, 11.5\} = 12.$$

The holder exercises. The continuation value belongs to a different policy.

Final synthesis

$$\boxed{\text{American pricing} = \text{no-arbitrage valuation} + \text{an optimal stopping decision.}}$$

Dynamic programming, the Snell envelope, and the free-boundary PDE are three compatible ways to represent the same exercise right.